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VLSI Design and Testing(BEC602)

CMOS Process Technology

1 Silicon Semiconductor Technology

1.1 Introduction

- Silicon, in its pure (intrinsic) state, is a semiconductor with an electrical resistance that lies between that of a conductor and an insulator.
- Unlike metals, whose conductivity is primarily due to free electrons, silicon's electrical properties can be significantly modified by introducing impurities into its crystal lattice.
- The process of adding impurity atoms to silicon to alter its conductivity is called **doping**.
- The introduced atoms, known as **dopants**, can either donate free electrons or create electron vacancies (holes)
- Certain impurity elements accept electrons from the silicon lattice, creating **holes** as the majority charge carriers.
- These dopants are called **acceptors**, and common examples include boron (B) and gallium (Ga).
- Other impurity elements contribute free **electrons** to the silicon lattice, making electrons the majority charge carriers.
- These dopants are called **donors**, with common examples being phosphorus (P) and arsenic (As).
- When **n-type** and **p-type** silicon materials are brought together, a transition region known as a **PN junction** is formed.
- This junction plays a crucial role in semiconductor devices.
- By arranging multiple PN junctions in specific physical structures, various semiconductor devices such as diodes, transistors, and integrated circuits (ICs) can be designed.
- Semiconductor fabrication has evolved significantly, allowing precise control over junction properties, leading to highly efficient and miniaturized electronic components.

1.2 Wafer Processing

- Semiconductor manufacturing begins with **silicon wafers**, thin disks of single-crystal silicon, typically 75 mm to 150 mm in diameter and less than 1 mm thick.

- These wafers are cut from ingots grown using the **Czochralski method**, the most common technique for producing high-purity silicon.

1.2.1 Czochralski Method

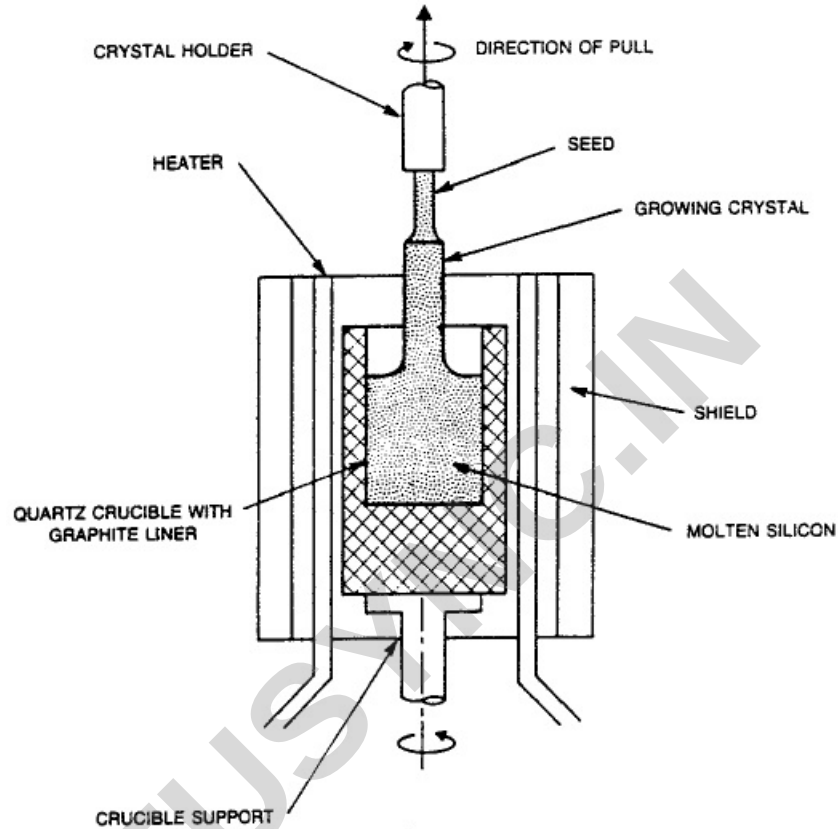


Figure 1: Czochralski process for manufacturing silicon ingots

- High-purity polycrystalline silicon is melted in a **quartz crucible**, surrounded by a **graphite radiator**, and heated using **radio frequency induction**.
- The temperature is maintained slightly above 1425°C , with an **inert gas** (helium or argon) preventing contamination.
- A **seed crystal** is dipped into the melt and slowly **withdrawn while rotating**.
- The molten silicon solidifies, maintaining the single-crystal structure of the seed.
- The **ingot diameter** is controlled by the **withdrawal rate** and **rotation speed**, with growth rates ranging from 30 mm h^{-1} to 180 mm h^{-1} .

1.2.2 Wafer Slicing and Polishing

- The grown ingot is sliced into wafers using **diamond-tipped blades**, with thicknesses between 0.25 mm and 1.0 mm , depending on diameter.
- At least one face is **polished to a mirror finish** to remove surface irregularities and ensure a defect-free surface for semiconductor fabrication.

1.3 Oxidation

- Silicon dioxide (SiO_2) plays a crucial role in the fabrication of silicon integrated circuits.
- Many manufacturing techniques depend on its insulating and passivating properties, making reliable SiO_2 formation essential.

1.3.1 Oxidation Process

- Silicon oxidation is achieved by heating silicon wafers in an oxidizing atmosphere, such as oxygen or water vapor.
- The process consumes silicon and results in an SiO_2 layer that expands in both vertical directions.
- The two common oxidation methods are:

1. **Wet Oxidation:**

- Uses an oxidizing atmosphere containing **water vapor**.
- Operates at 900°C to 1000°C .
- Provides a **faster oxidation rate**.

2. **Dry Oxidation:**

- Uses **pure oxygen** as the oxidizing agent.
- Requires higher temperatures (1200°C) for acceptable growth rates.
- Produces a **denser and higher-quality** oxide layer.

1.3.2 Oxide Growth

- Since SiO_2 has roughly **twice the volume** of the consumed silicon, the oxide layer grows almost equally above and below the original silicon surface.
- This effect is illustrated in **n-channel MOS devices**, where the field oxide projects in both directions.

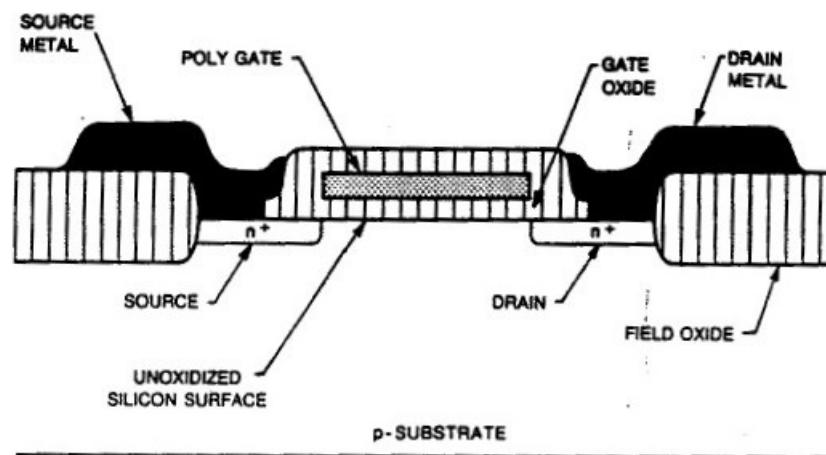


Figure 2: An nMOS transistor showing the growth of field oxide in both vertical directions

1.4 Selective Diffusion

- Selective diffusion is a process used to introduce controlled amounts of **donor or acceptor impurities** into specific areas of a silicon wafer.
- The ability of silicon dioxide (SiO_2) to act as a diffusion barrier allows precise patterning of doped regions.
- Selective diffusion involves the following steps:
 - **Opening windows** in an SiO_2 layer grown on the wafer surface.
 - **Removing SiO_2 selectively** using a suitable **etching** technique.
 - **Exposing the silicon** to a controlled **dopant source** to alter its electrical properties.

1.4.1 Photolithography and Etching

The removal of SiO_2 is achieved using a process called **photolithography**, which involves:

- Coating the wafer with a **photosensitive resist (PR)**.
- Exposing PR to **ultraviolet (UV) light** through a patterned mask.
- Developing the PR to remove unpolymerized areas.
- Etching away the exposed SiO_2 , allowing selective diffusion of dopants.

Conventional UV-based lithography limits feature sizes to about $1.5\ \mu\text{m}$ to $2\ \mu\text{m}$ due to diffraction effects.

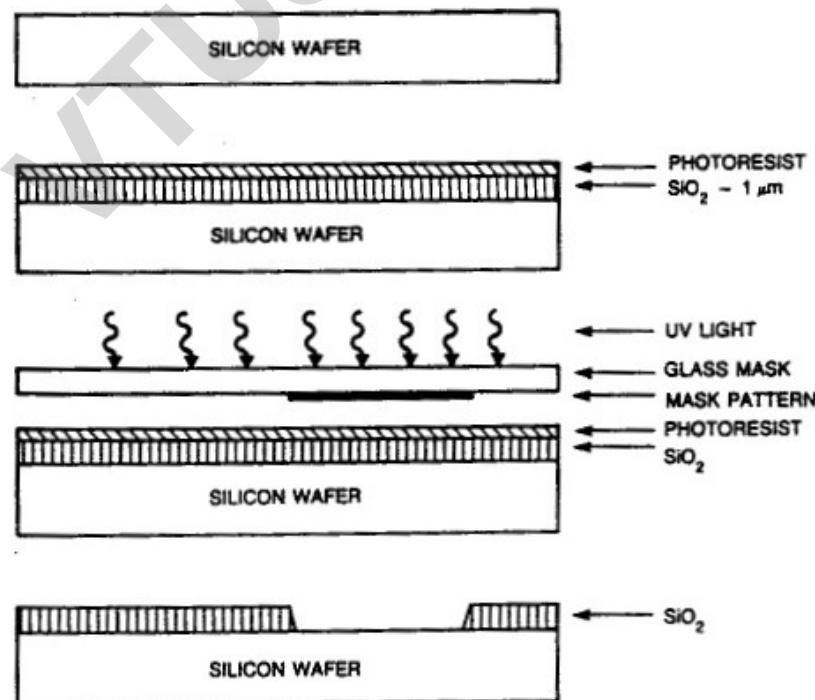


Figure 3: Simplified steps involved in the patterning of SiO_2

1.4.2 Electron Beam Lithography (EBL)

Electron beam lithography (EBL) has emerged as a more precise technique, achieving line widths of approximately $0.5\text{ }\mu\text{m}$. Key advantages of EBL include:

- **Direct patterning** from digital data without masks.
- **High flexibility**, allowing different patterns in different wafer regions.
- **Quick modifications** to design layouts.

However, EBL has limitations, including **high equipment costs** and **long processing times**, which restrict its commercial adoption.

1.4.3 Conclusion

Selective diffusion, enabled by SiO_2 masking, is a crucial step in semiconductor fabrication. While traditional photolithography remains dominant, **EBL offers higher resolution** at the cost of speed and expense.

1.5 The Silicon Gate Process

- In addition to single-crystal silicon and silicon dioxide (SiO_2), **polysilicon** (polycrystalline silicon) plays a crucial role in semiconductor fabrication.
- It is used as:
 - An **interconnect** in integrated circuits.
 - The **gate electrode** in MOS transistors.
- The self-aligned nature of polysilicon gates enables precise definition of source and drain regions, improving circuit performance.

1.5.1 Steps in the Silicon Gate Process

The silicon gate process involves multiple cycles of **photomasking** and **oxide etching**, as illustrated in figure below.

Step 1: Field Oxide Formation

A thick layer of SiO_2 (field oxide) is grown over the wafer surface. This oxide is selectively **etched** in areas where transistors will be placed.

Step 2: Gate Oxide Growth

A **thin, controlled** layer of SiO_2 , called the **gate oxide**, is grown on the exposed silicon surface. This serves as the insulating layer beneath the polysilicon gate.

Step 3: Polysilicon Deposition

A layer of **undoped polysilicon** is deposited across the wafer. This layer is later **etched** to form transistor gates and interconnections.

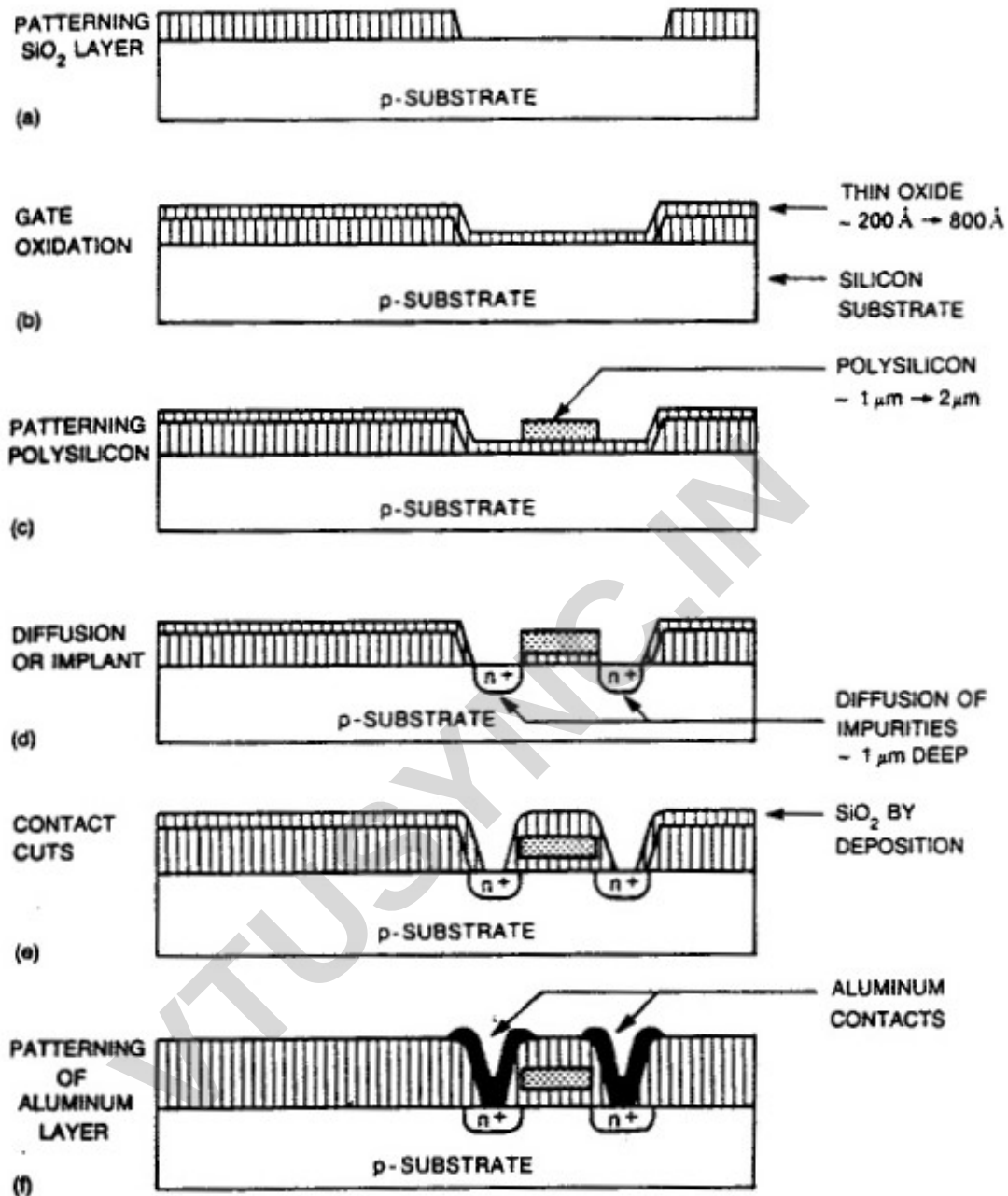


Figure 4: Fabrication steps for a silicon gate nMOS transistor

Step 4: Dopant Diffusion

The wafer is exposed to a dopant source, leading to:

- Formation of **source and drain junctions** in the silicon substrate.
- **Doping of polysilicon**, reducing its resistivity.

The polysilicon gate acts as a mask, ensuring that the source and drain regions do not extend beneath it. This **self-aligned** process enhances device performance.

Step 5: Final Processing

- The structure is covered with an SiO_2 layer.
- **Contact holes** are etched for electrical connections.
- A metallic layer (e.g., **aluminum**) is deposited and etched to complete the circuit.

1.5.2 Conclusion

- The silicon gate process enables precise control over MOS transistor fabrication.
- The use of polysilicon as a gate material allows for **self-aligned** diffusion, reducing overlap capacitance and improving circuit speed.

2 CMOS Technologies

2.1 The p-well process

- CMOS (Complementary Metal-Oxide-Semiconductor) technology is a fundamental building block in modern VLSI (Very Large-Scale Integration) circuits.
- Among the various CMOS fabrication techniques, the **p-well process** is a widely used method.
- In this approach, a **moderately doped n-type substrate** (wafer) is used as the starting material.
- A **p-type well** is created in this substrate to accommodate the **n-channel MOS transistors**, while the native n-type substrate is used for fabricating **p-channel MOS transistors**.
- This process allows for the integration of both n-channel and p-channel transistors on the same wafer, enabling complementary circuit operation.
- The fabrication of CMOS devices involves multiple steps, including photolithography, diffusion, ion implantation, oxidation, and metallization.

Step 1: P-Well Formation

- The first step involves defining the p-well region using a photolithographic mask.
- The field oxide (FOX) is etched away in the designated p-well regions.
- A deep diffusion or ion implantation process introduces p-type dopants (typically boron) into the n-type substrate.
- The depth and concentration of the p-well must be carefully controlled, as they influence the threshold voltage and breakdown voltage of the nMOS transistors.

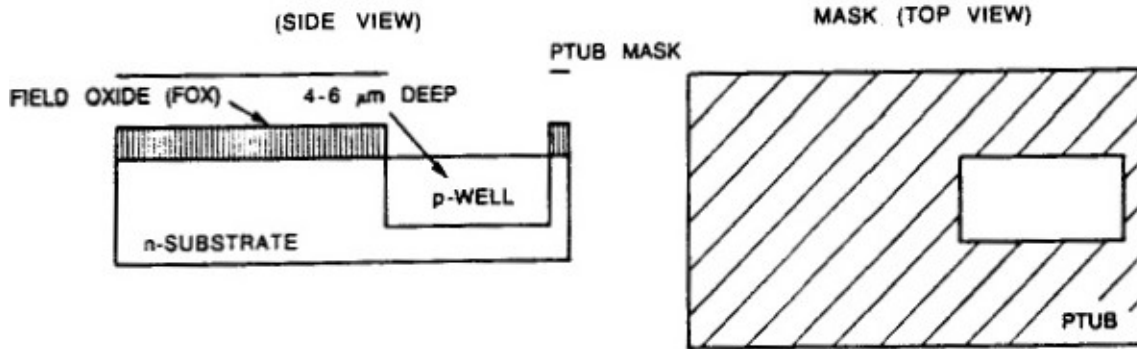


Figure 5: P-Well Formation

Step 2: Thin Oxide Growth

- A thin, high-quality silicon dioxide (SiO_2) layer is thermally grown on the exposed silicon surface.
- This oxide layer acts as the gate insulator for MOS transistors.
- The thickness of this layer is crucial for transistor performance and determines the gate capacitance.

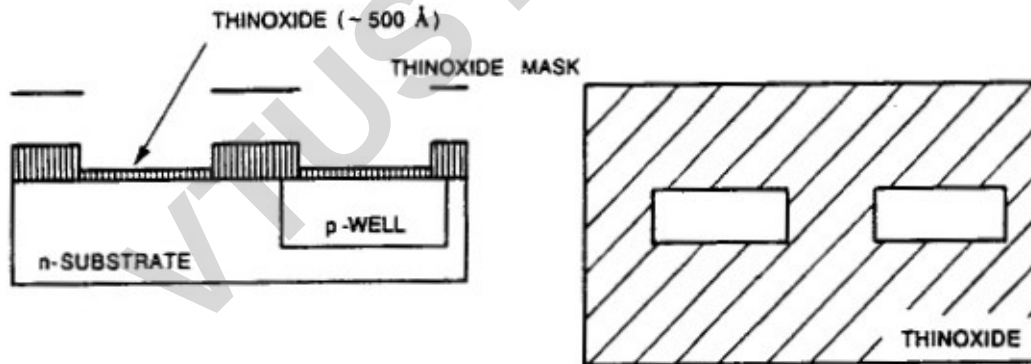


Figure 6: Thin Oxide Growth

Step 3: Polysilicon Gate Formation

- A layer of polycrystalline silicon (polysilicon) is deposited over the entire wafer.
- This layer serves as the gate electrode for MOS transistors.
- Using photolithography and etching, the polysilicon is patterned to define transistor gates and interconnections.

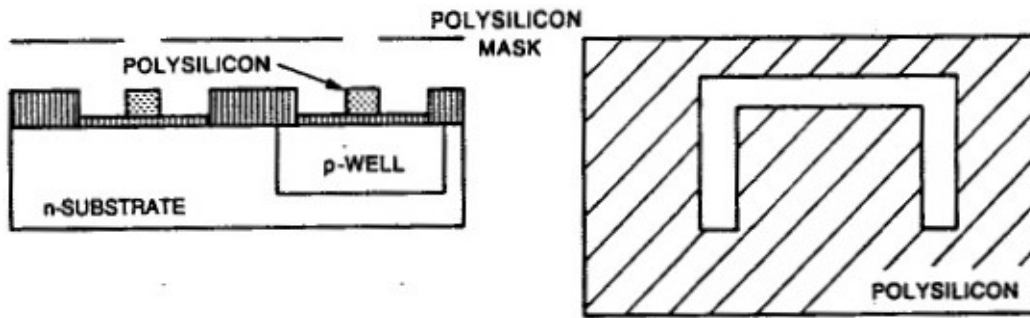


Figure 7: Polysilicon Gate Formation

Step 4: Source and Drain Implantation

- A **p-plus (p^+)** mask is used to define regions where p-type dopants will be implanted.
- In the absence of the p^+ mask, the remaining thin oxide areas are doped with n-type impurities, forming n^+ regions.
- These diffusion regions serve as the source and drain terminals for nMOS and pMOS transistors.
- The polysilicon gate acts as a mask, ensuring precise alignment of source and drain regions, which is known as the **self-aligned process**.

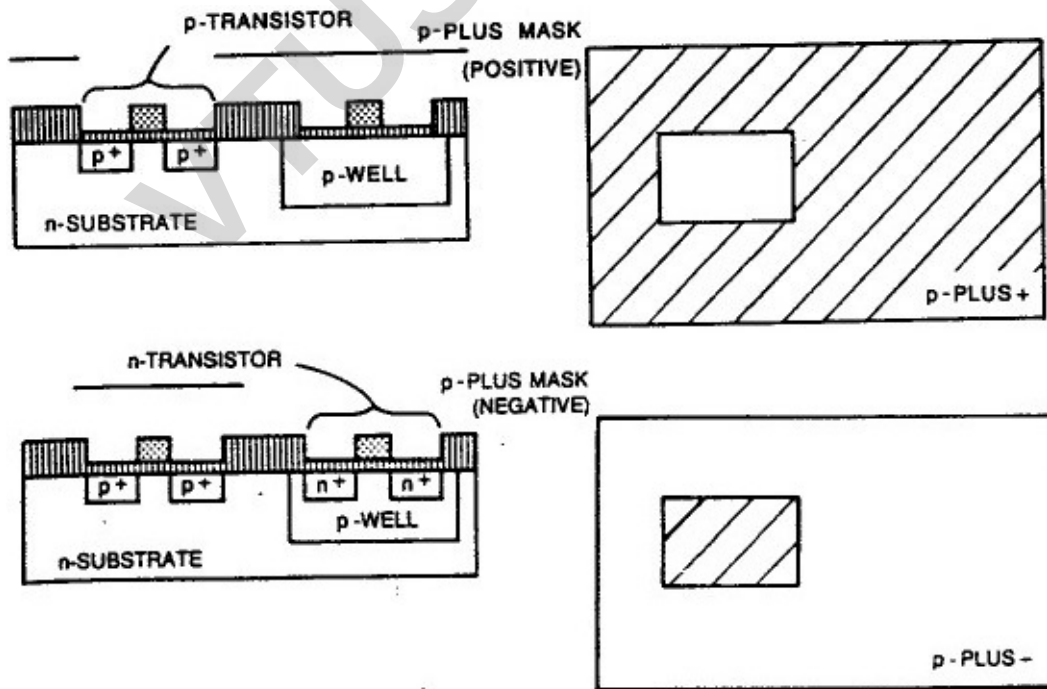


Figure 8: Source and Drain Implantation

Step 5: Oxide Deposition and Contact Formation

- A thick insulating layer of silicon dioxide (SiO_2) is deposited over the entire wafer.
- Photolithography is used to etch contact holes in the oxide layer, exposing selected diffusion and polysilicon regions.

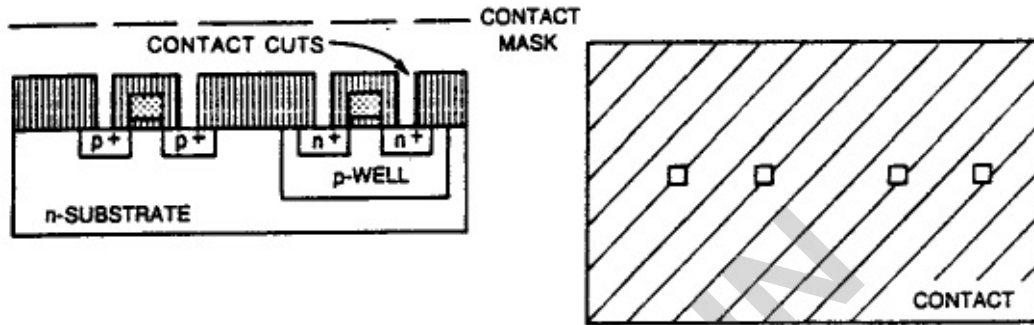


Figure 9: Oxide Deposition and Contact Formation

Step 6: Metallization

- A thin layer of aluminum or another conductive metal is deposited over the wafer.
- The metal layer is patterned and etched to form interconnections between various circuit elements.

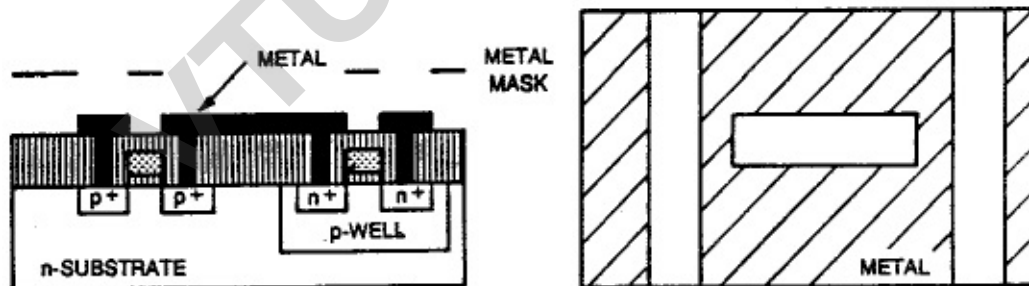


Figure 10: Metallization

Step 7: Passivation and Final Processing

- A protective passivation layer (such as silicon nitride) is deposited to shield the circuit from contamination and environmental damage.
- Openings are etched to allow external connections to bonding pads.

Substrate and Well Connections

Proper biasing of the substrate and well is essential to ensure correct circuit operation and prevent unwanted effects like **latch-up**. The connections are as follows:

- The **p-well** is connected to the negative supply voltage (V_{SS} or ground).
- The **n-substrate** is connected to the positive supply voltage (V_{DD}).
- Substrate contacts help minimize parasitic resistance and improve circuit reliability.

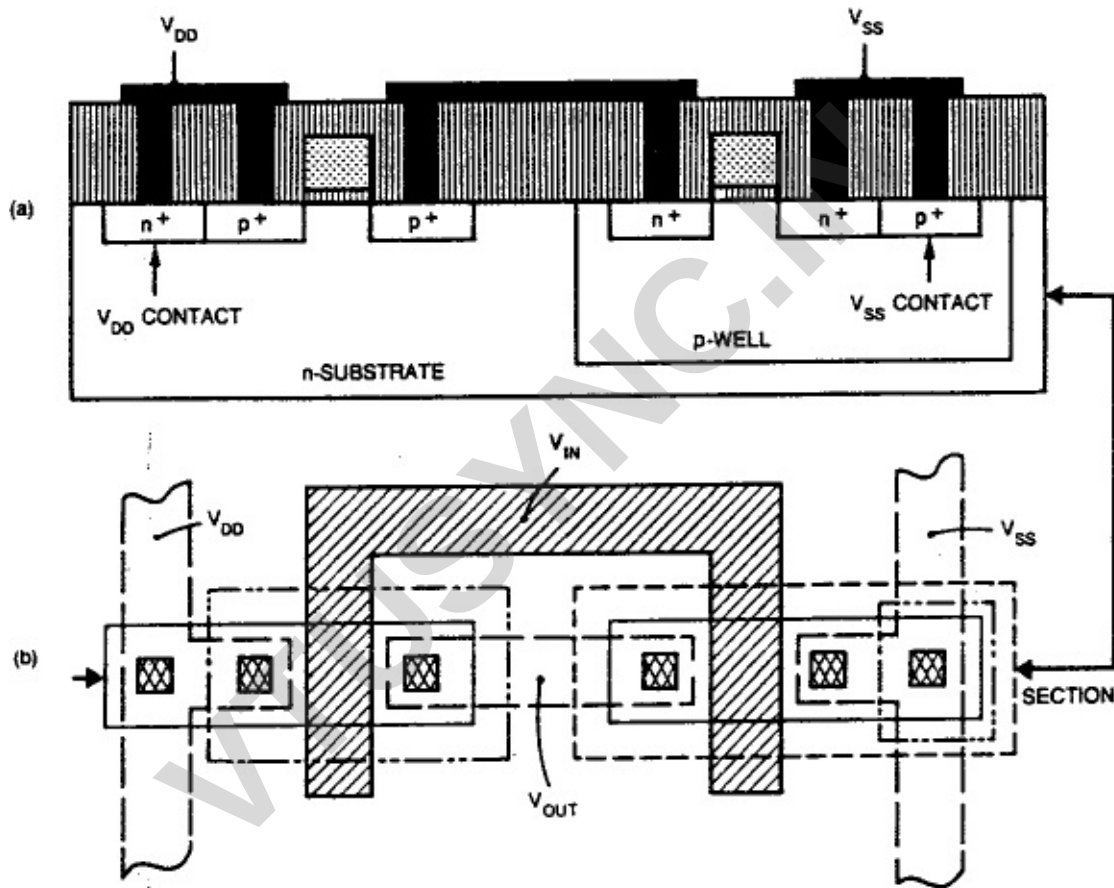


Figure 11: p-well substrate contacts: (a) Substrate and Well Connections, (b) Corresponding Layout

2.1.1 Advanced P-Well Processes

1. Retrograde P-Well Process

To overcome limitations of conventional p-well diffusion, an advanced technique called the **retrograde p-well process** is used. In this method:

- High-energy boron implantation is used instead of thermal diffusion.
- The implanted dopants remain at a controlled depth, reducing lateral diffusion.
- This enables smaller transistor spacing, improving packing density.

- Junction depth, sheet resistance, and threshold voltage can be independently controlled.

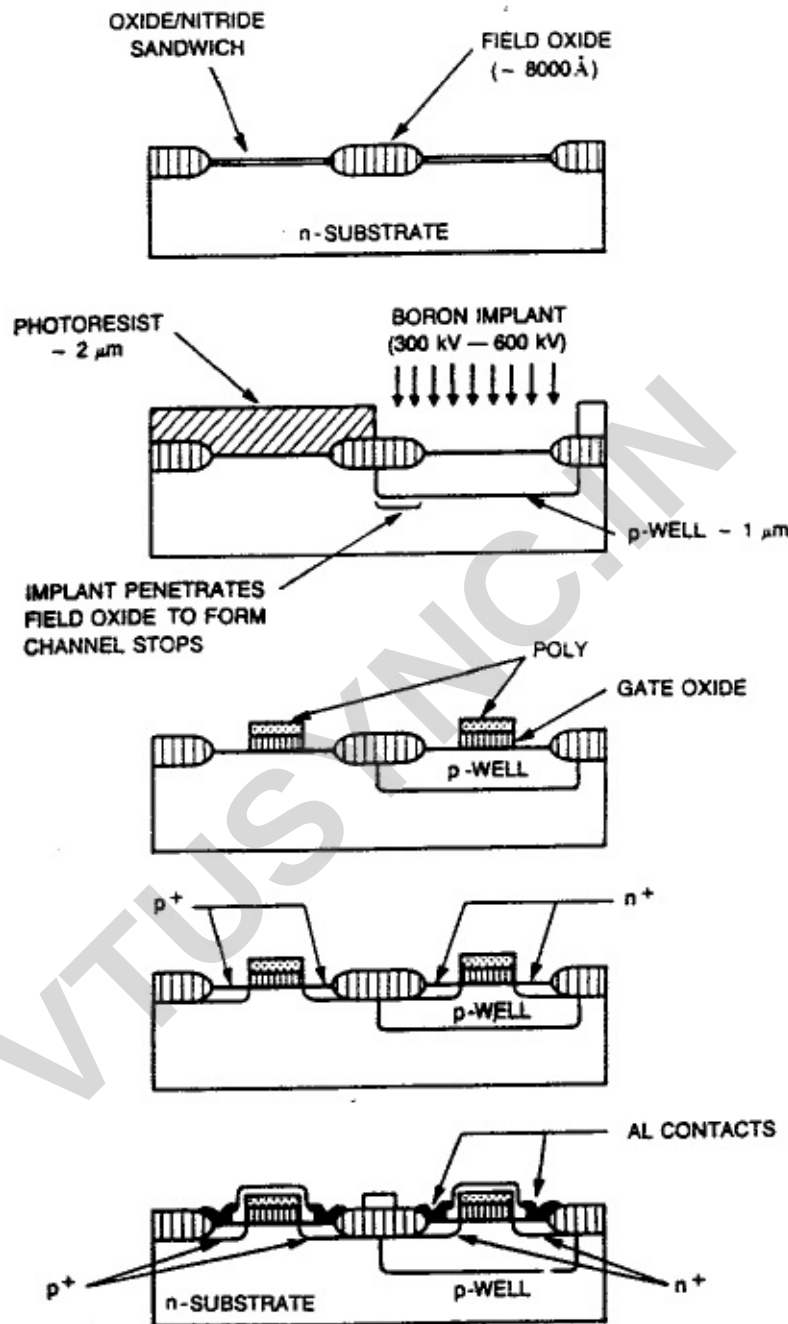


Figure 12: GE-Intersil's "retrograde p-well" process

2. CMOSC Process

The **CMOSC process** is another enhancement of the p-well CMOS technology. Key features include:

- Boron implants define p-channel transistors.

- Phosphorus implants define n-channel transistors.
- Improved control over lateral diffusion reduces leakage currents.
- Enhanced field oxide processing minimizes thinning effects (such as *bird's beak*).

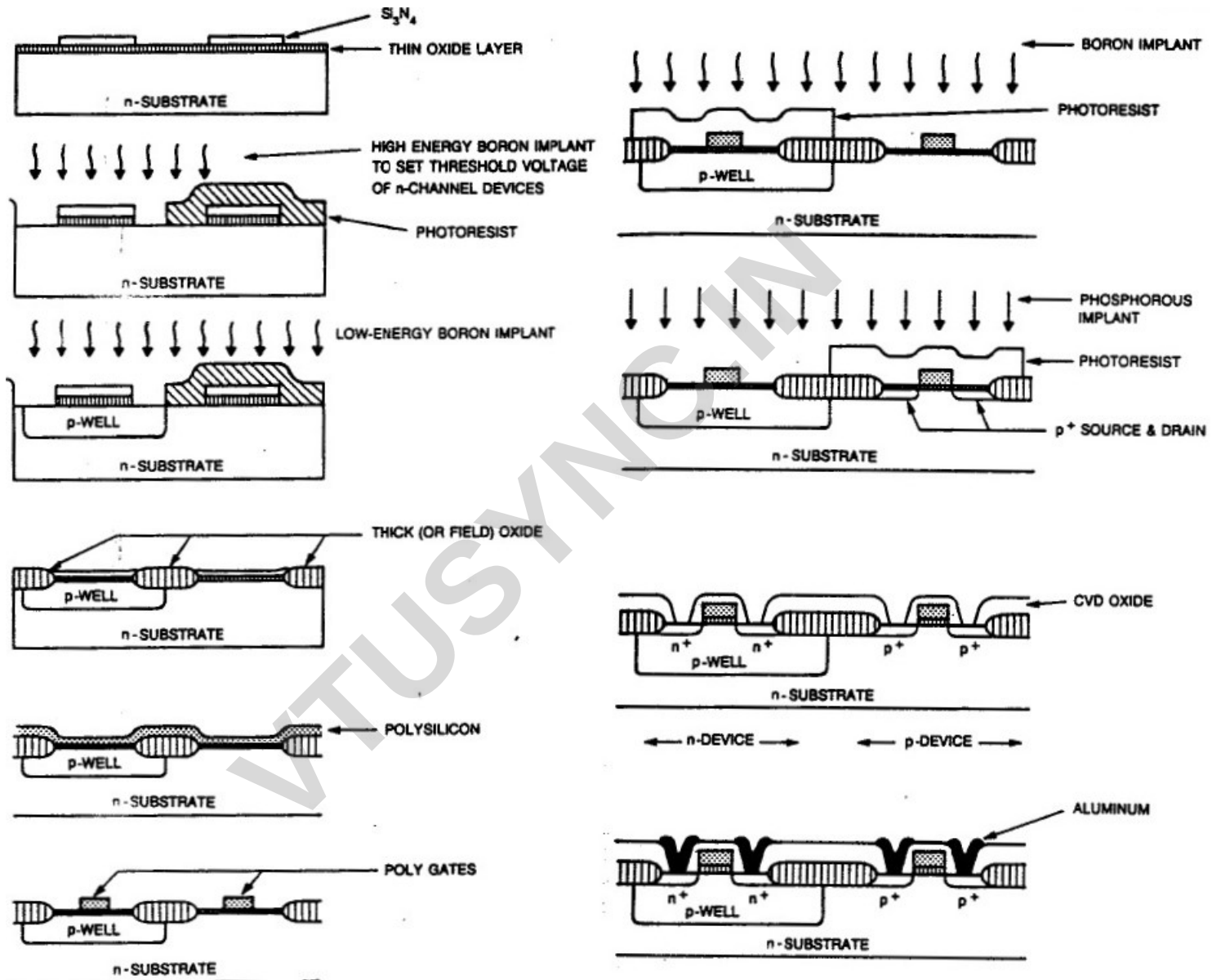


Figure 13: Hewlett- Packard's CMOSC process

2.1.2 Challenges and Considerations

While the p-well CMOS process offers a practical approach for integrating nMOS and pMOS transistors, it has some challenges:

- **Threshold Voltage Control:** P-well doping concentration must be carefully optimized to achieve the desired threshold voltages.

- **Parasitic Capacitance:** The n-transistors in the p-well experience higher source/drain capacitance, which can affect switching speed.
- **Latch-Up Prevention:** Proper well connections and substrate contacts are necessary to prevent latch-up, which can cause circuit failure.
- **Area Considerations:** The required spacing between p- and n-transistors can increase chip area, affecting integration density.

2.2 The n-well process

- The **n-well process** is one of the most commonly used CMOS fabrication techniques.
- It has largely replaced the p-well process due to its compatibility with existing nMOS process lines, allowing manufacturers to transition from nMOS to CMOS technology efficiently.
- The key advantage of the n-well process is that it enables the integration of both n-channel and p-channel MOSFETs on a p-type substrate.
- The pMOS transistors are placed within the n-well region, ensuring better noise immunity and reducing substrate bias effects compared to a single-well approach.

2.2.1 Fabrication Steps of the n-Well CMOS Process

Step 1: Substrate Preparation

- A p-type silicon substrate is used as the base material.
- A thin oxide layer is grown to protect the silicon surface.

Step 2: n-Well Formation

- An n-well mask defines the well regions.
- A phosphorus implant introduces n-type dopants.
- A high-temperature diffusion step ensures proper well depth and uniformity.

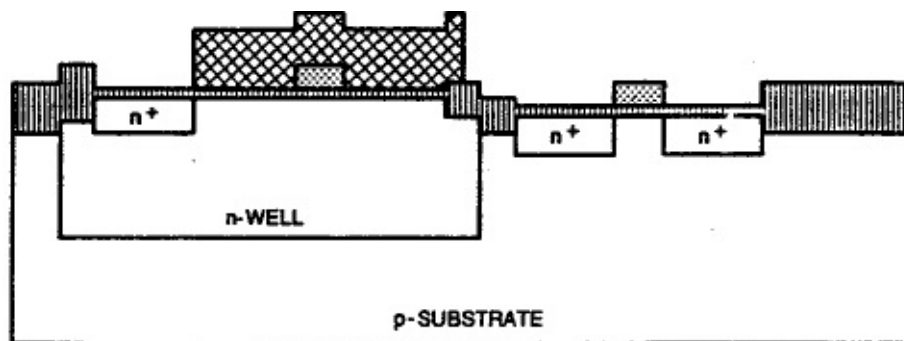


Figure 14: n-Well Formation

Step 3: Active Area Definition

- Field oxide is removed from transistor regions.
- The oxide thickness is adjusted for device isolation.

Step 4: Threshold Voltage Adjustment

- A light p-type doping step adjusts the threshold voltage of transistors.

Step 5: Polysilicon Gate Formation

- A thin oxide layer is grown as gate oxide.
- Polysilicon is deposited, patterned, and etched to form gates.

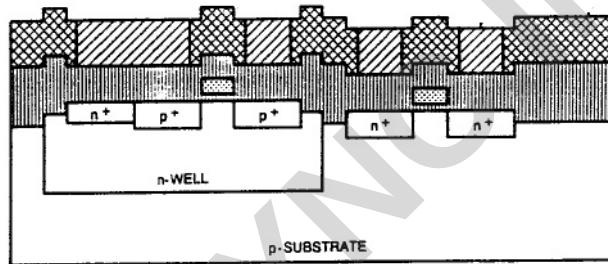


Figure 15: Polysilicon Gate Formation

Step 6: Source and Drain Formation

- Arsenic (n-type) is implanted for n-channel source and drain.
- Boron (p-type) is implanted for p-channel source and drain.

Step 7: Contact and Metallization

- Contact openings are etched to expose terminals.
- Aluminum is deposited and patterned for interconnections.

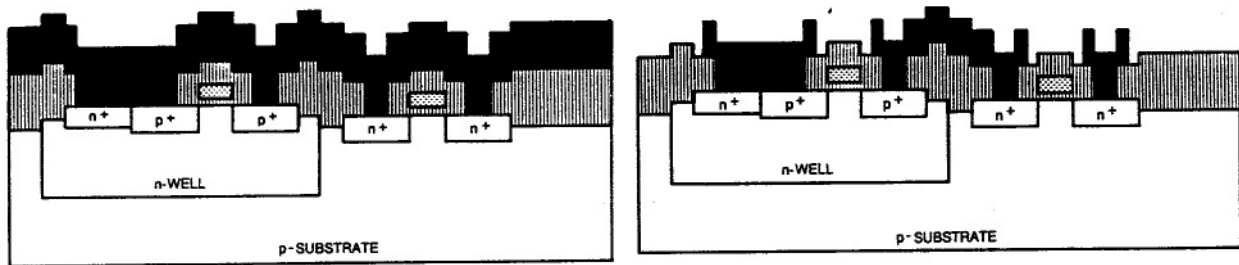


Figure 16: Contact and Metallization

2.2.2 Key Characteristics of the n-Well Process

- **pMOS Transistors in n-Well**

- p-channel transistors are isolated within the n-well, preventing unwanted interactions with the p-substrate.
- However, pMOS devices suffer from increased body effect, requiring careful design.

- **nMOS Transistors in p-Substrate**

- n-channel transistors are directly fabricated in the p-type substrate.
- They exhibit higher electron mobility, leading to faster switching speeds.

2.2.3 Comparison of n-Well and p-Well CMOS Processes

Table 1: Comparison of n-Well and p-Well CMOS Processes

Feature	n-Well CMOS	p-Well CMOS
Substrate Type	p-Type	n-Type
Well Type	n-Well	p-Well
pMOS Location	Inside n-Well	Directly in n-Substrate
nMOS Location	Directly in p-Substrate	Inside p-Well
Advantages	Compatible with nMOS processes, better noise isolation	Lower junction capacitance for pMOS
Disadvantages	Higher body effect for pMOS	Requires additional processing for nMOS

2.2.4 The Berkeley n-Well Process

- The Berkeley n-well CMOS process is often described using the **Process Input Description Language (PIDL)**, which defines key fabrication steps.
- Some important commands used in PIDL include:

Table 2: Process Input Description Language commands.

Command	Function
SUBSTRATE	Defines the substrate type and impurity level.
OXIDE	Specifies oxide thickness.
DEPOSITION	Deposits material layers.
ETCH	Removes material to a specified depth.
DOPE	Introduces dopants to control electrical properties.
MASK	Defines patterned regions for processing.

For instance, polysilicon gate formation involves:

1. Depositing 0.3 μm of polysilicon: `DEPO POLY THICK=0.30.`
2. Applying a resist layer: `DEPO RST THICK=0.5.`

3. Using a mask to define the gate: MASK RST DRST MSI POSI.
4. Etching the exposed polysilicon: ETCH POLY DEPTH=0.6.
5. Removing the resist, leaving the polysilicon gate.

2.3 The Twin-Tub Process

2.3.1 Introduction

- CMOS technology has evolved through various fabrication techniques to improve performance and design flexibility.
- One such advancement is the **Twin-Well** or **Twin-Tub CMOS Process**, which enables independent optimization of both p-type and n-type transistors.
- This process allows the threshold voltage, body effect, and gain associated with n-channel and p-channel transistors to be individually optimized, leading to improved circuit performance.

2.3.2 Key Features of the Twin-Well Process

The Twin-Tub process offers several advantages over the conventional p-well process:

- **Independent Optimization:** The twin-tub approach allows separate control of p-well and n-well properties, leading to better transistor performance.
- **Threshold Voltage Control:** The use of separate wells makes it possible to fine-tune the threshold voltages of both p-type and n-type MOSFETs.
- **Improved Body Effect and Gain:** The body effect, which influences the threshold voltage, can be optimized separately for nMOS and pMOS transistors.
- **Reduced Latch-up Susceptibility:** The process starts with an epitaxial silicon layer to enhance latch-up immunity by isolating the wells from the substrate.

2.3.3 Process Sequence

- The fabrication sequence of the Twin-Tub CMOS process is similar to the p-well process, with the key difference being that both p-well and n-well regions are formed independently to achieve the desired electrical properties.
- The key steps are as follows:

1. Substrate Preparation:

- A lightly doped silicon wafer (typically p-type) is used as the starting material.
- A thin oxide layer is grown to serve as a protective layer.
- An epitaxial silicon layer is deposited to improve performance and minimize latch-up issues.

2. Twin-Tub Formation:

- Both p-well and n-well are created by selective ion implantation.
- The process allows separate optimization of the electrical properties of p-channel and n-channel MOSFETs.

3. Threshold Voltage Adjustment:

- A threshold adjustment implant is introduced to control the transistor threshold voltages.

4. Field Oxidation and Isolation:

- Field oxide is grown to isolate different transistor regions.
- LOCOS (Local Oxidation of Silicon) may be used to create field oxide regions.

5. Polysilicon Deposition and Gate Formation:

- A thin gate oxide layer is thermally grown.
- Polysilicon is deposited and patterned to form the gate electrode.

6. Source and Drain Implantation:

- Arsenic (n-type) is implanted into the p-well to form nMOS source and drain.
- Boron (p-type) is implanted into the n-well to form pMOS source and drain.

7. Contact Cut Definition:

- Openings are etched to allow metal contacts to connect to the source, drain, and gate terminals.

8. Metallization:

- Metal layers (usually aluminum) are deposited and patterned to create interconnections.
- A passivation layer is deposited for protection.

2.3.4 Advantages of the Twin-Tub Process

- **Independent Optimization:** The electrical characteristics of nMOS and pMOS transistors can be separately optimized, unlike the p-well or n-well processes.
- **Enhanced Performance:** Since both n-well and p-well are present, better control of parameters such as threshold voltage, body effect, and gain is possible.
- **Improved Latch-up Immunity:** The epitaxial layer provides additional isolation, reducing latch-up issues.

2.3.5 Illustration of the Twin-Tub Process

The figure below illustrates the process steps involved in the AT&T-Bell Laboratories twin-tub process.

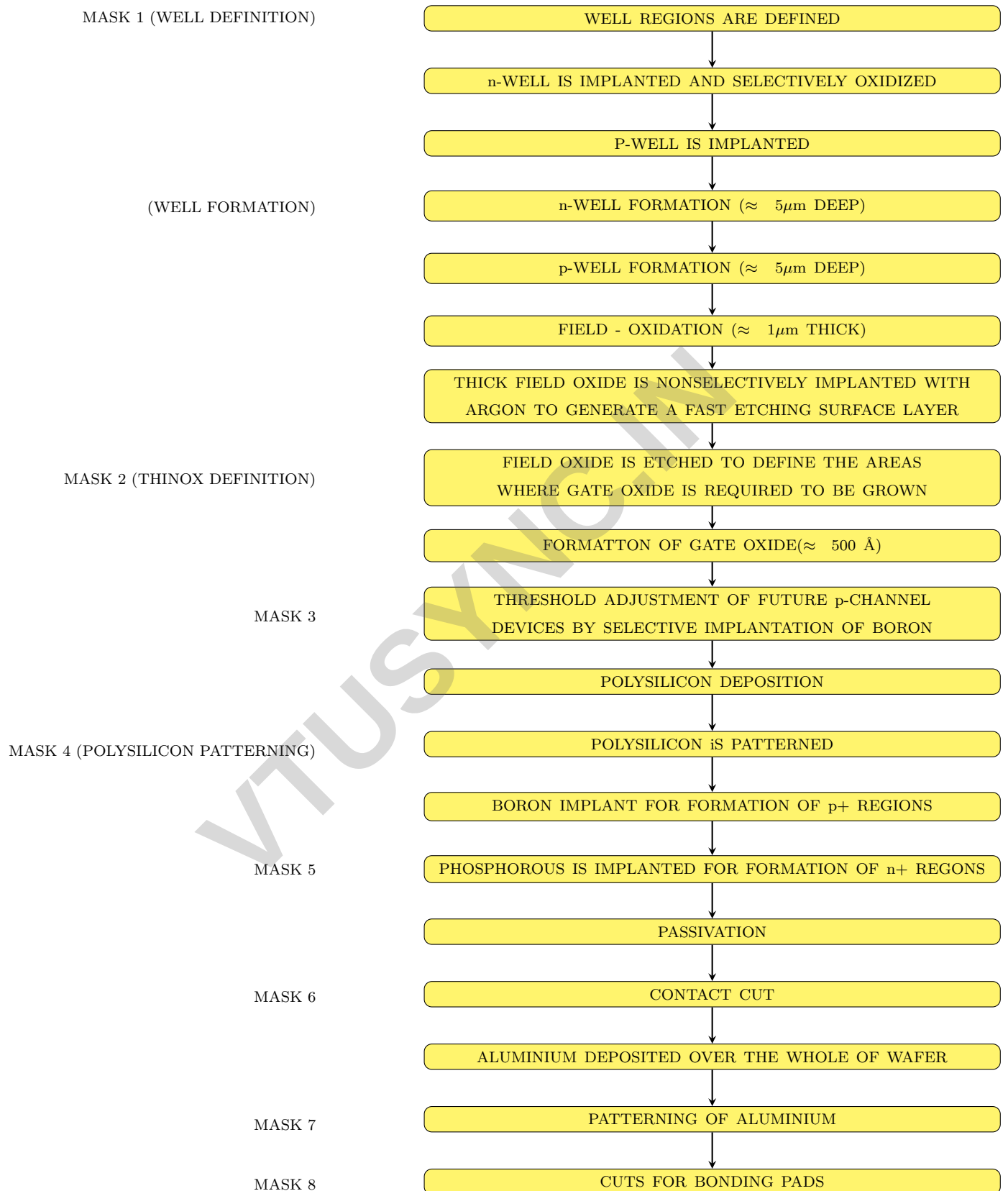


Figure 17: AT&T Bell Laboratories' twin-tub CMOS process steps

2.3.6 Final Structure

The figure below presents the final twin-tub CMOS structure, highlighting the independently optimized p-well and n-well regions, including the required substrate contacts.

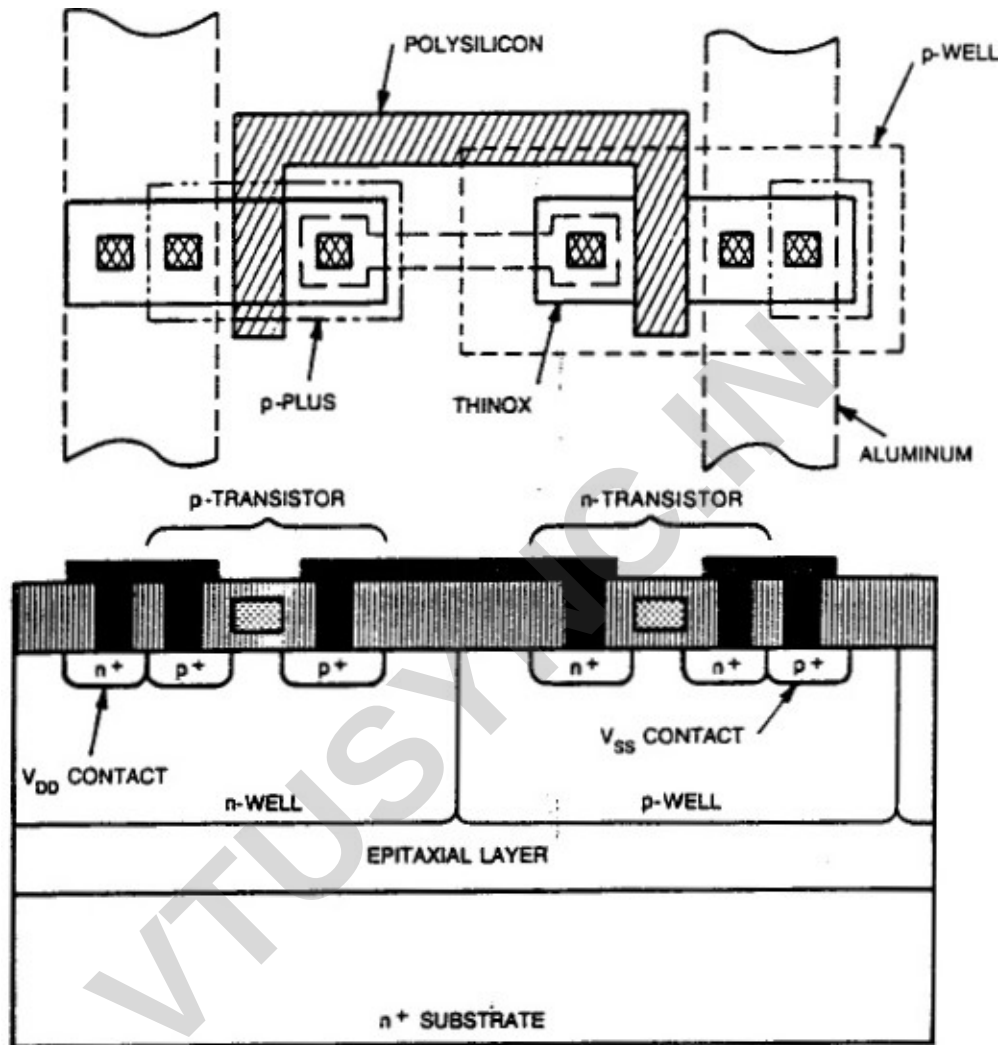


Figure 18: Twin-tub process cross-section and layout of an inverter

2.3.7 Conclusion

- The twin-tub CMOS process offers significant advantages over the conventional p-well process.
- The independent formation of p-well and n-well allows better control over transistor parameters such as threshold voltage, gain, and body effect.
- Moreover, the inclusion of an epitaxial layer enhances the process by improving latch-up immunity, making the twin-tub process ideal for high-performance CMOS applications.

2.4 Silicon on Insulator (SOI) CMOS Technology

2.4.1 Introduction

- Silicon on Insulator (SOI) CMOS technology is an advanced fabrication process that offers several advantages over traditional bulk silicon CMOS technologies.
- These advantages include higher device density, elimination of latch-up problems, and reduced parasitic capacitances.

2.4.2 SOI CMOS Process Steps

1. Substrate Preparation:

- A thin layer (7–8 μm) of lightly doped n-type silicon is epitaxially grown on an insulating substrate, such as sapphire.

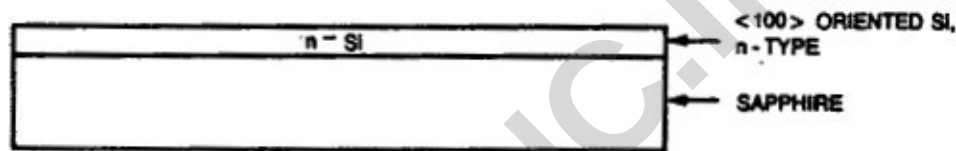


Figure 19: Substrate Preparation

2. Island Formation:

- An anisotropic etch is used to remove silicon except in areas where diffusion regions (n or p) are needed.
- This creates isolated silicon “islands” for device formation.

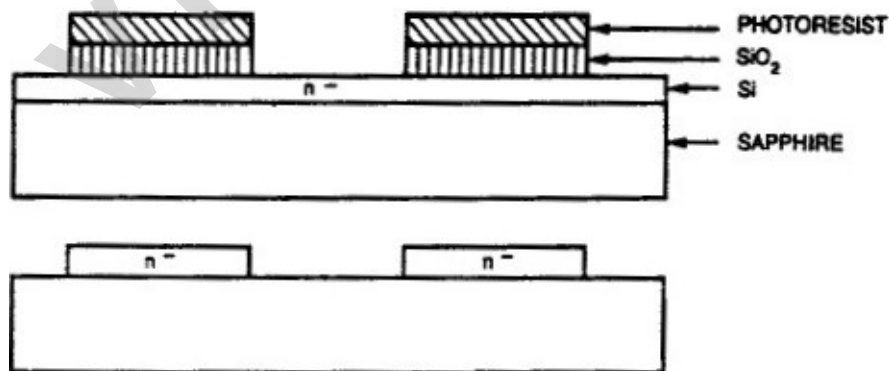


Figure 20: Island Formation

3. Doping for p-Islands:

- The n-islands are masked with photoresist, and a p-type dopant (e.g., boron) is implanted to form p-islands.
- These p-islands will become the n-channel devices.

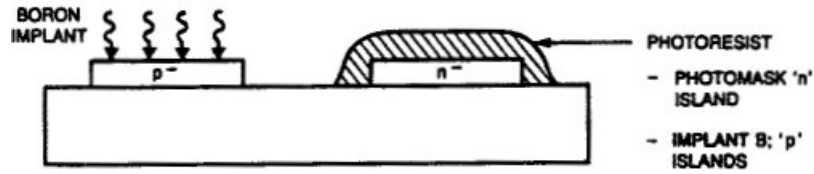


Figure 21: Doping for p-Islands

4. Doping for n-Islands:

- The p-islands are masked, and an n-type dopant (e.g., phosphorus) is implanted to form n-islands.
- These n-islands will become the p-channel devices.

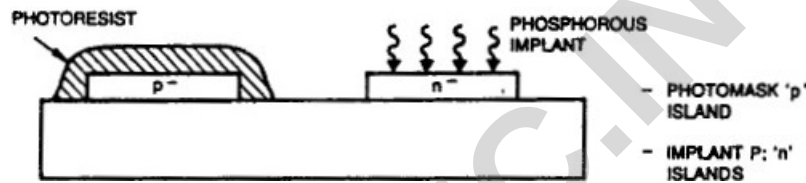


Figure 22: Doping for n-Islands

5. Gate Oxide Formation:

- A thin gate oxide (500–600 Å) is grown over the silicon structures using thermal oxidation.

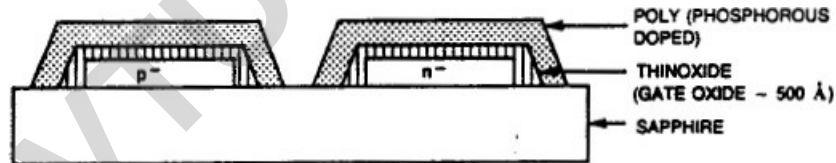


Figure 23: Gate Oxide Formation

6. Polysilicon Deposition:

- A polysilicon film is deposited over the oxide and doped with phosphorus to reduce resistivity.
- The polysilicon is then patterned and etched to define the gate regions.



Figure 24: Polysilicon Deposition

7. Source/Drain Formation for n-Channel Devices:

- The n-islands are masked, and an n-type dopant (e.g., phosphorus) is implanted to form the source and drain regions of the n-channel devices.
- The polysilicon gate blocks dopant implantation in the gate region.

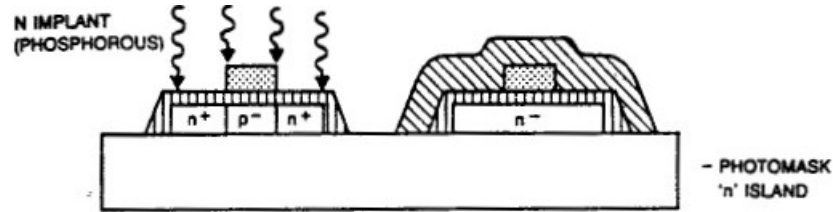


Figure 25: Source/Drain Formation for n-Channel Devices

8. Source/Drain Formation for p-Channel Devices:

- The p-islands are masked, and a p-type dopant (e.g., boron) is implanted to form the source and drain regions of the p-channel devices.
- The polysilicon gate blocks dopant implantation in the gate region.

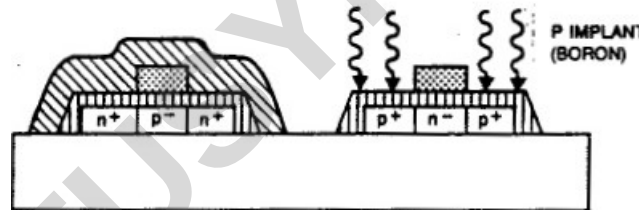


Figure 26: Source/Drain Formation for p-Channel Devices

9. Insulator Deposition and Metallization:

- A layer of phosphorus glass or silicon dioxide is deposited over the structure.
- Contact cuts are etched, and aluminum is evaporated and patterned to form interconnects.

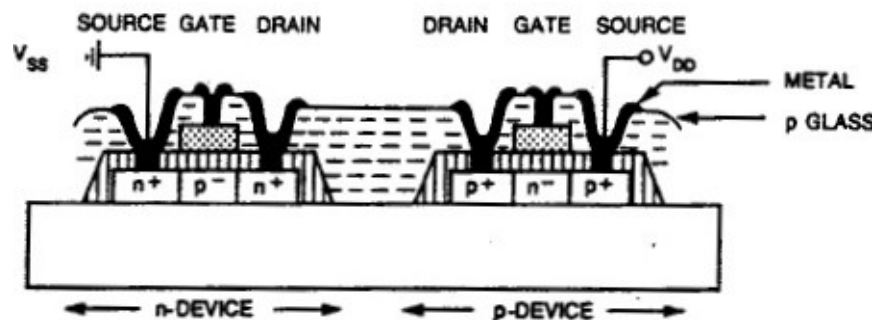


Figure 27: Insulator Deposition and Metallization

10. Passivation:

- A final passivation layer of phosphorus glass is deposited and etched over bonding pad locations.

2.4.3 Advantages of SOI Technology

- **Higher Density:** The absence of wells allows for denser device structures compared to bulk silicon. Direct n-to-p connections are also possible.
- **Lower Parasitic Capacitances:** The insulating substrate reduces parasitic junction capacitances, enabling faster circuit operation.
- **No Latch-Up:** The isolation of n- and p-transistors by the insulating substrate eliminates latch-up issues.
- **No Field-Inversion Problems:** The insulating substrate prevents field-inversion problems common in bulk silicon.
- **No Body Effect:** The absence of a conducting substrate eliminates body effect problems.
- **Enhanced Radiation Tolerance:** SOI devices are more resistant to radiation, making them suitable for aerospace and high-radiation environments.

2.4.4 Challenges of SOI Technology

- **Input Protection:** The absence of substrate diodes makes input protection more challenging.
- **Lower Device Gains:** Device gains are lower, requiring larger I/O structures.
- **Higher Cost:** Single crystal sapphire or spinel substrates are significantly more expensive than silicon.
- **Less Developed Processing Techniques:** SOI processing techniques are less mature compared to bulk silicon, leading to higher complexity and cost.
- **Preferential Etch Requirements:** To improve yield, preferential etching is often used to taper island edges, adding complexity to the process.

2.4.5 Comparison of Etch Techniques

- **Isotropic Etch:** Creates rounded sidewalls but is less precise for dense structures.
- **Anisotropic Etch:** Produces vertical sidewalls with no undercut, but can create high step heights for metal or polysilicon runners.
- **Preferential Etch:** Tapers island edges to minimize step heights, improving yield and ease of metallization.

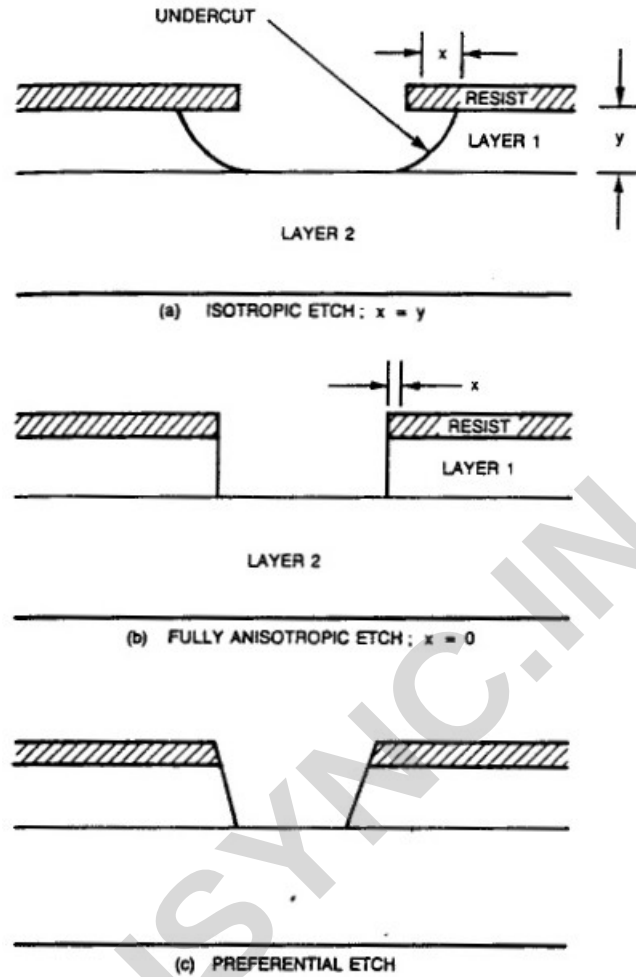


Figure 28: Classification of etching process

2.4.6 Conclusion

- SOI CMOS technology offers significant performance advantages, including higher speed, lower power consumption, and enhanced radiation tolerance.
- However, its higher cost and processing complexity have limited its widespread adoption.
- Despite these challenges, SOI remains a promising technology for high-performance and specialized applications, such as aerospace, military, and high-speed computing.

3 Layout Design Rules

3.1 Introduction

- Layout design rules, also known as design rules, serve as guidelines for preparing the photomasks used in the fabrication of integrated circuits (ICs).
- These rules act as a critical communication bridge between the **circuit designer** and the **process engineer**, ensuring that the manufacturing process is both efficient and reliable.

- The primary objective of layout design rules is to **achieve the highest possible yield** while maintaining a **compact geometry** without compromising circuit reliability. A trade-off exists:
 - **Conservative rules** $\Rightarrow$ Higher reliability but may limit performance.
 - **Aggressive rules** $\Rightarrow$ Potential performance gains but may reduce fabrication yield.

3.1.1 Purpose of Design Rules

Design rules define geometric constraints for layout artwork, ensuring that the processed wafer maintains the intended **topology and geometry** of the design. It is important to note:

- Design rules **do not represent a strict boundary** between correct and incorrect fabrication.
- A layout that violates some design rules may still function, but frequent violations increase the risk of fabrication failures.

3.1.2 Key Design Rule Constraints

Two critical aspects of design rules include:

1. Line Widths:

- **Too narrow:** May cause discontinuities, leading to circuit failure.
- **Too close:** Can result in **short circuits** between adjacent conductive paths.

2. Interlayer Registration:

- Ensures proper alignment between different material layers in the fabrication process.
- Incorrect spacing between layers can cause **electrical and manufacturing defects**.

3.1.3 Types of Design Rules

Several approaches exist for defining design rules:

1. Micron-Based Rules:

- Specifies **minimum feature sizes** and **spacings** in absolute micron values (e.g., minimum thinox width = $4\ \mu\text{m}$).
- Commonly used in industrial fabrication processes.

2. Alpha (α) and Beta (β) Rules:

- Uses β as a reference for feature sizes and α as the minimum required grid spacing.
- These parameters are related by a constant factor.

3. Lambda (λ)-Based Rules:

- Introduced by **Mead and Conway**, lambda-based rules use a single parameter (λ) to define all feature sizes.
- Enables **scaling of circuit layouts** across different technology nodes.

- Allows rules to be expressed compactly, often on a single page.
- However, **λ -based scaling is only an approximation** and may not be suitable for commercial circuits.

3.1.4 Derivation of Lambda-Based Rules

A representative example of how **lambda (λ)-based rules** are derived from micron-based rules is shown in table below.

Table 3: Derivation of Lambda-Based Rules from Micron Rules

Mask	Feature	Micron Rule (μm)	Lambda Rule (λ)
Thinox	Minimum thinox width	$4\mu\text{m}$	2λ
	Minimum thinox spacing	$4\mu\text{m}$	2λ
	Minimum p-thinox to n-thinox spacing	$8\mu\text{m}$	4λ
Polysilicon	Minimum poly width	$3.75\mu\text{m}$	2λ
	Minimum poly spacing	$3.75\mu\text{m}$	2λ
	Minimum gate poly width (p)	$4.5\mu\text{m}$	3λ
	Minimum gate poly width (n)	$4\mu\text{m}$	2λ
	Minimum gate poly extension	$3.5\mu\text{m}$	2λ
Aluminum	Minimum Al width	$4.5\mu\text{m}$	3λ
	Minimum Al spacing	$4.5\mu\text{m}$	3λ

3.1.5 Practical Considerations

- **Deviating from design rules** may not always lead to failure, but frequent violations increase the likelihood of fabrication issues.
- While **λ -rules** offer a simplified abstraction, modern **symbolic techniques** allow designers to work without worrying about the specifics of design rules.

3.2 Layer Representations in CMOS Processes

3.2.1 Introduction

- The advances in CMOS fabrication processes have made them increasingly complex, making it difficult to visualize all the mask levels involved in actual IC fabrication.
- However, the design process can be abstracted into a manageable number of **conceptual layout levels**, which represent the physical features observed in the final silicon wafer.
- At a sufficiently high conceptual level, all CMOS processes include the following features:
 - Two different substrates
 - Doped regions forming both **p**- and **n**-transistors

- Transistor gate electrodes
- Interconnection paths
- Interlayer contacts

3.2.2 Layer Representations

Different representation techniques are used to depict various layers in CMOS processes. These techniques include:

- A **color scheme** proposed by JPL
- A **modified color scheme** distinguishing between nMOS and CMOS structures
- **Stipple patterns**
- **Line styles**
- A **combination** of the above

Some of these representations are illustrated in following tables.

1. JPL and Mead-Conway Layer Representation

Table 4: JPL/Mead-Conway Layer Representation for p-well CMOS Process

Layer	JPL Color	Symbolic Representation	Comments
p-well	Brown	-	Inside brown is p-well, outside is n-type substrate
Thin oxide	Green	n-transistor	Thin oxide may not cross a well boundary
Poly	Red	Polysilicon	Generally n
p	Yellow	p-transistor	Inside is p
Metal1	Light blue	Metal1	-
Metal2	Dark blue	Metal2	-
Contact cut	Black	Contact	-
Overglass	-	-	-

2. Alternate Layer Representations

Table 5: Alternate Layer Representations for p-well CMOS Process

Layer	Alternate Color	CIF Code
p-well	Brown	CW
Thin oxide	Red	CD
Poly	Green	CP
p	Purple	CS
Metal1	Tan	CM
Metal2	Dark blue	CN
Contact	Black	CC
Overglass	white stipples	CG

3. CIF Representation of Process Descriptions

- The CIF (Caltech Intermediate Form) layer names used by JPL for bulk CMOS are presented in table below.
- CIF version 2.0 uses up to four alphanumeric characters to describe a mask level.
- The first letter represents the process class (e.g., 'C' for bulk CMOS), followed by a second character identifying the layer type.

Table 6: First Character Representation of CIF for Process Description

Process	CIF Character
n-channel MOS	N
p-channel MOS	P
Bulk CMOS processes	C
Silicon-on-Insulator (SOI) processes	S

Mask Levels in CMOS and SOI Processes

- In n-well bulk CMOS, the only difference from p-well CMOS is the **reversal of well and original substrate roles**.
- Different process lines may use different combinations of **n-well**, **p-well**, or **twin-tub masks** to define the process.
- Understanding the specific set of masks used in a process is crucial when designing interface formats.
- For example, an **n-mask** (reverse of a p-mask) may be used to denote n-transistors.

SOI vs. Bulk CMOS

- The mask levels in Silicon-on-Insulator (SOI) processes are conceptually simpler.
- The visible geometry corresponds directly to the design features, such as **n-regions** and **p-regions**.
- The key difference between SOI and bulk CMOS is the **absence of wells** in SOI.

The Role of Symbolic Design

- To simplify bulk CMOS design, symbolic-level representations are encouraged.
- This approach allows designers to manipulate **n- and p-transistors** directly without worrying about specific design rules.
- The benefits of symbolic design include:
 - Eliminating the need for explicit layout design rules.
 - Enabling the next generation of **automatically performance-optimized circuits**.

3.3 Lambda-based p-well rules

To standardize layout design, lambda-based rules define:

- Minimum feature sizes.
- Spacing requirements.
- Overlap constraints.

Table below summarizes the lambda-based layout rules for the p-well process.

Table 7: Lambda-based layout rules

NO.	MASK	FEATURE	DIMENSION
1	Thinox	A1. Minimum thinox width	2λ
		A2. Minimum thinox spacing n^+/p^+ to n^+/p^+	2λ
		A3. Minimum p-thinox to n-thinox spacing	8λ
2	p-well	B1. Minimum p-well width	4λ
		B2. Minimum p-well spacing (wells at same potential)	2λ
		B3. Minimum p-well spacing (wells at different potential)	6λ
		B4. Minimum distance to internal thinox	3λ
		B5. Minimum distance to external thinox	5λ
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NO.	MASK	FEATURE	DIMENSION
3	Poly	C1. Minimum poly width	2λ
		C2. Minimum poly spacing	2λ
		C3. Minimum poly to thinox spacing	λ
		C4. Minimum poly gate extension	2λ
		C5. Minimum thinox source/drain extension	2λ
4	p-plus	D1. Minimum overlap of thinox	$1.5 - 2\lambda$
		D2. Minimum p-plus spacing	2λ
		D3. Minimum gate overlap or distance to gate edge	$1.5 - 2\lambda$
		D4. Minimum spacing to unrelated thinox	$1.5 - 2\lambda$
5	Contact	E1. Minimum contact area	$2\lambda \times 2\lambda$
		E2. Minimum contact to contact spacing	2λ
		E3. Minimum overlap of thinox or poly over contact	λ
		E4. Minimum spacing to gate poly	2λ
		E5. n- source/drain contact	
		E6. p- source/drain contact	
		E7. V_{SS} contact	
		E8. V_{DD} contact	
		E9. Split contact V_{SS}	
		E10. Split contact V_{DD}	
6	Metal	F1. Minimum metal width	$2 - 3 \lambda$
		F2. Minimum metal spacing	3λ
		F3. Minimum metal overlap of contact	λ

Note: For E5 - E10, See the pictorial representation shown in following section.

3.3.1 Pictorial Representations

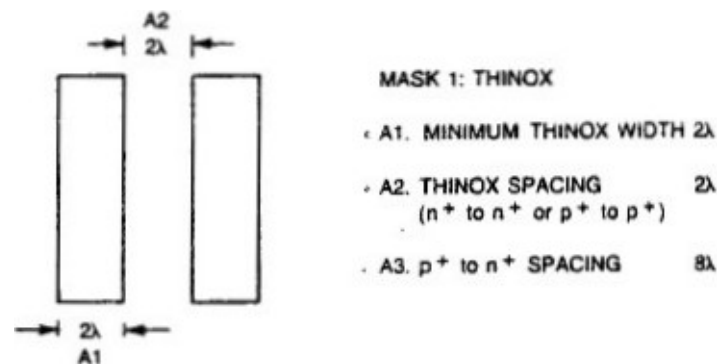


Figure 29: Mask 1: Thinox

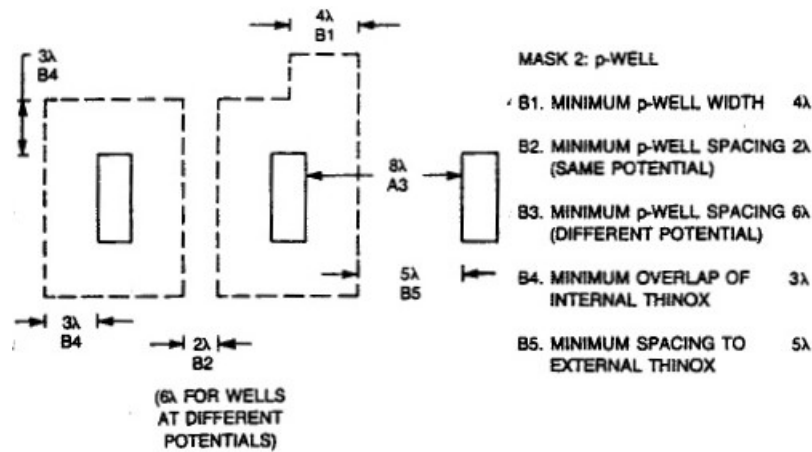


Figure 30: Mask 2: p-well

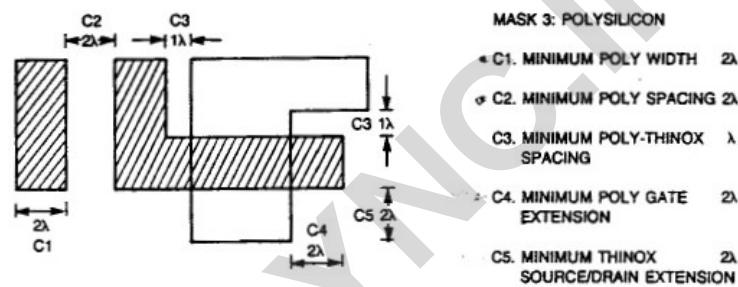


Figure 31: Mask 3: Polysilicon

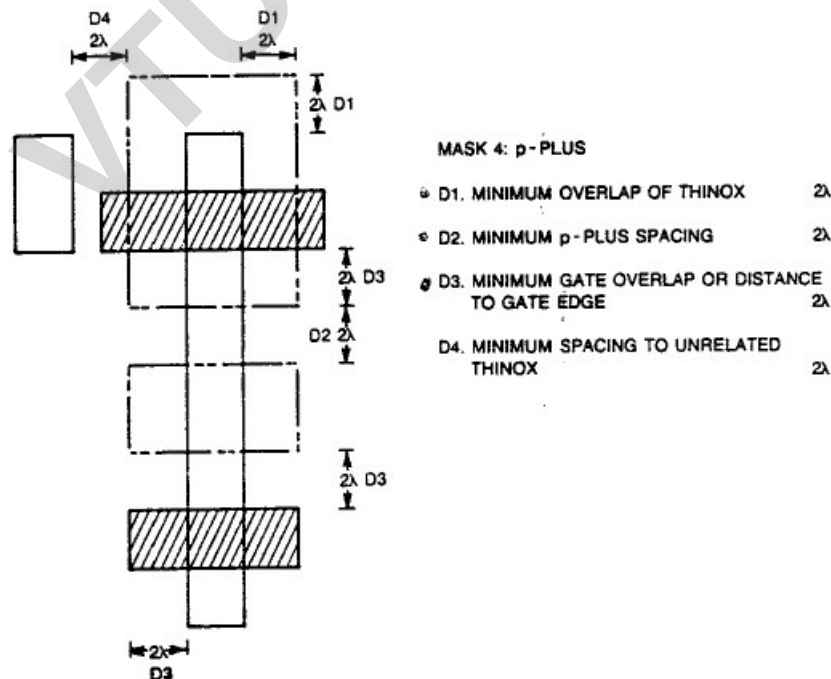


Figure 32: Mask 4: p - plus

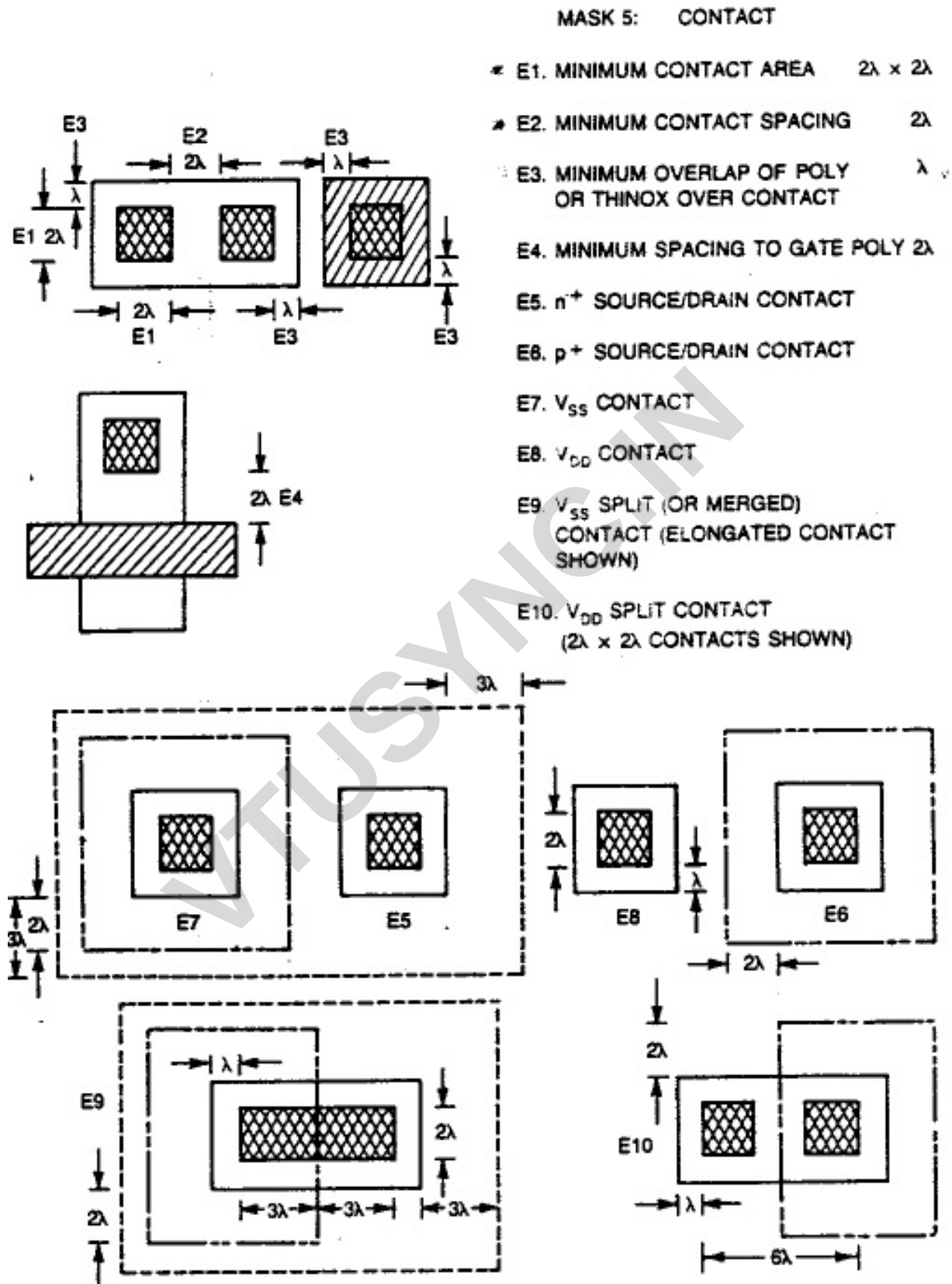


Figure 33: Mask 5: Contact

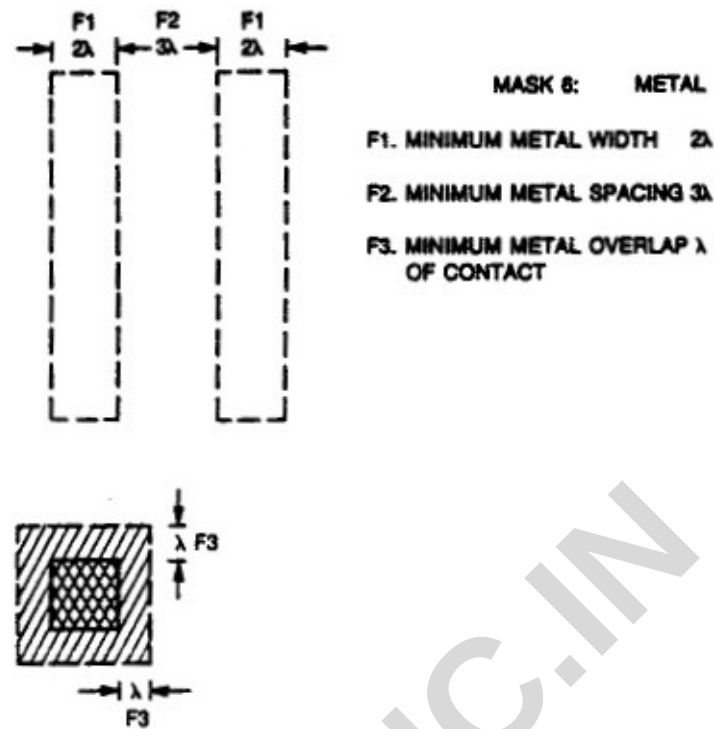


Figure 34: Mask 6: Metal

3.3.2 Key Considerations

1. Well Spacing and Separation

- The p-well is a deep diffusion requiring sufficient spacing to prevent electrical shorts.
- The inside clearance must account for *field oxide transitions* across the well boundary.
- The **thin oxide layer should not cross a well boundary** to prevent electrical shorts.

2. Transistor Layout Rules

- The source and drain diffusion is self-aligned to the gate using the poly region.
- The **gate extension** must be at least **$1.5-2\lambda$ beyond the diffusion edge**.
- Thin oxide regions not forming a transistor should be kept separated by **$0.5-1\lambda$** .

3. Contact Design in CMOS Layout

There are several types of contacts:

- Metal-to-diffusion (n or p).
- Metal-to-polysilicon.
- V_{DD} . and V_{SS} . substrate contacts.

- Split (merged) contacts for better conductivity.

Each well must be tied to the correct supply voltage:

- p-well $\rightarrow V_{SS}$.
- n-well $\rightarrow V_{DD}$.

4. Poly Doping Considerations

- In p-well CMOS processes, poly is usually doped n-type.
- Doping variations can increase poly sheet resistance.

5. Guard Rings for Noise Isolation

- Guard rings collect injected minority carriers.
- **p+ guard rings** in the **n-substrate** are tied to V_{SS} .
- **n+ guard rings** in the **p-well** are tied to V_{DD} .

3.4 Lambda-Based SOI Rules

- Table below presents a set of lambda rules for CMOS SOI technology.
- An interesting aspect of these rules is that, apart from a few exceptions such as the **n-device to n-device spacing rule, implant rules, metal-to-metal spacing rule, and contact rules**, most of the layout constraints follow a consistent scaling factor.
- The primary value that designers need to keep in mind is 2λ .
- One significant application of the 2λ spacing rule is to ensure sufficient separation between the **island edge and unrelated polysilicon**.
- This rule is crucial in preventing electrical shorts between polysilicon and island edges, which could occur due to **thin or faulty oxide coverage over the islands**.

Table 8: Lambda-Based Layout Rules for SOI

NO.	MASK	FEATURE	DIMENSION
1	Island	Minimum island width	2λ
		p-device to n-device spacing	2λ
		n-device to n-device spacing	3λ
		p-device to p-device spacing	3λ
2	Implant	Implant/island overlap	λ
		Implant/island spacing	λ
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NO.	MASK	FEATURE	DIMENSION
3	Poly	Minimum poly width	2λ
		Minimum poly-poly spacing	2λ
		Minimum poly to island	2λ
		Minimum island edge to poly spacing	2λ
		Minimum poly extension over island	2λ
4	Contact	Distance over poly edge	λ
		Distance over island edge	λ
		Distance from island edge	λ
		Distance from non-contacted feature	2λ
		Contact width on island	2λ
		Contact width on poly	2λ
5	Metal	Minimum metal width	3λ
		Minimum metal spacing	2λ
		Minimum metal overlap of contact	λ

- These lambda-based rules offer a systematic approach to SOI layout design, ensuring proper electrical isolation and manufacturability.
- The 2λ rule for poly-to-island edge separation is especially important to prevent electrical shorts due to fabrication imperfections.

3.4.1 Pictorial Representation

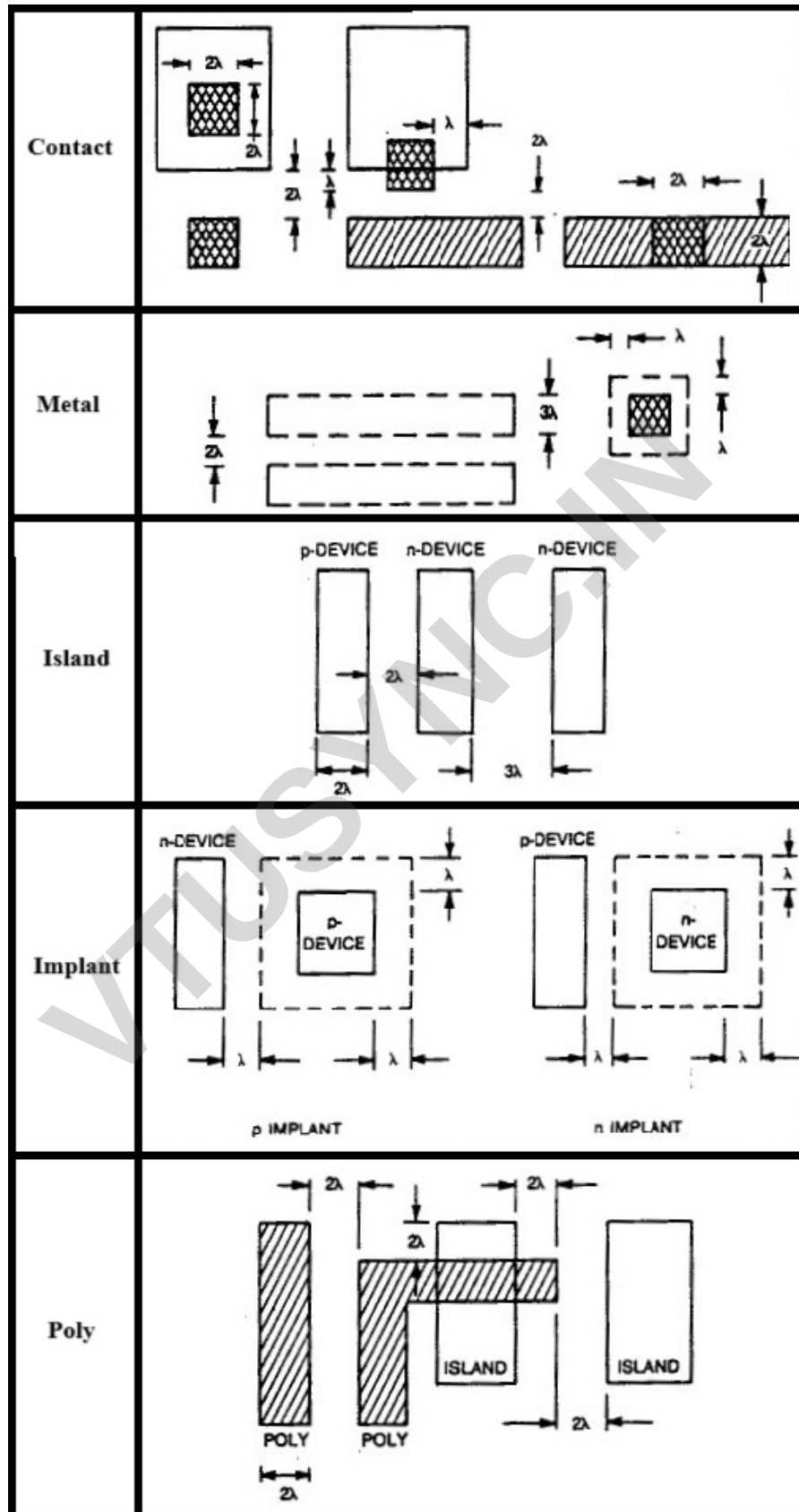


Figure 35: Lambda - based rules for SOI

3.5 Double Metal Design Rules

- In modern VLSI design, the use of multiple metal layers improves routing efficiency and enhances signal integrity.
- The second metal layer (Metal2) works in conjunction with the first metal layer (Metal1) to provide additional connectivity.
- However, specific design rules must be followed to prevent issues such as broken conductors or shorts due to the vertical topology.
- Tables below present typical rules for the interaction between Metal2 and Metal1 in a two-layer metal process.
- The increased width and separation requirements for Metal2 ensure robustness against process variations, reducing the likelihood of open circuits and unintended shorts between adjacent wires.

Table 9: Double Metal Rules - Micron-Based (Suggested Lambda Rules)

Layer	Width	Spacing
Metal1	$3\ \mu\text{m}$ (2λ)	$4\ \mu\text{m}$ (3λ)
Metal2	$5\ \mu\text{m}$ (4λ)	$5\ \mu\text{m}$ (4λ)
Via	$3\ \mu\text{m} \times 3\ \mu\text{m}$ ($2\lambda \times 2\lambda$)	$3\ \mu\text{m}$ (2λ)
Cut	$3\ \mu\text{m} \times 3\ \mu\text{m}$ ($2\lambda \times 2\lambda$)	$3\ \mu\text{m}$ (2λ)
Cut-Via space	-	$3\ \mu\text{m}$ (2λ)

Table 10: Metal1, Metal2, and Via Constructs

Layer	Size
Metal 1	$5\ \mu\text{m} \times 5\ \mu\text{m}$ ($4\lambda \times 4\lambda$)
Via	$3\ \mu\text{m} \times 3\ \mu\text{m}$ ($2\lambda \times 2\lambda$)
Metal2	$7\ \mu\text{m} \times 7\ \mu\text{m}$ ($5\lambda \times 5\lambda$)

- The use of multiple metal layers enhances circuit performance and allows for higher routing densities.
- However, careful adherence to these design rules ensures that fabrication issues such as open circuits and short circuits are minimized.

3.6 Design Rules — Summary

- In commercial VLSI designs, lambda-based design rules are often insufficient to describe the complexities of high-performance circuits.

- Additional rules are required to ensure manufacturability and performance optimization.
- Some of the key additional rules found in advanced fabrication processes include:
 1. **Polysilicon Extension:** The extension of polysilicon in the direction that metal wires exit a contact.
 2. **Differing Gate Lengths:** Variations in gate lengths between p-transistors and n-transistors.
 3. **Gate Poly Extensions:** Different gate poly extension lengths depending on the device length or the device construction.
- While it is possible to worst-case these rules, doing so often results in inefficient designs.
- A more effective approach is to implement systems that automatically synthesize the correct geometry from an intermediate design representation.
- Symbolic design techniques provide a solution for developing generic CMOS circuits that can be adapted to a wide range of fabrication processes.
- Instead of relying solely on fixed design rules, these techniques allow for the abstraction of key design constraints and the automated generation of optimized layouts.